

CS560 - UI/UX Design & Implementation

8-1 Short Paper: Interface Strengths and Weaknesses

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LibraryStart Interface Strengths and Weaknesses Analysis

Usability testing is an important part of the user-centered design process because it helps designers understand how real users interact with a product. Instead of making assumptions about what users need, usability studies provide evidence that can be used to improve the design and create a better user experience. The LibraryStart usability study was conducted to evaluate how effectively incoming students could use an onboarding document designed to help them navigate the university library's online resources. The study focused on three major workflows: searching the library catalog, accessing research databases, and submitting interlibrary loan requests. Examining task completion rates, errors, completion times, and participant feedback, the library's UX team was able to identify both strengths and weaknesses in the onboarding document and the library system. According to Garofalo (2025), usability evaluation helps designers identify issues that may affect users' ability to complete tasks successfully and support continuous improvement through user-centered design.

Purpose of the Usability Study

The purpose of the usability study was to determine whether incoming students could efficiently and accurately complete common library tasks after reading the LibraryStart onboarding document. Twenty-four students participated in the study and completed nine tasks across three modules while researchers observed their behavior and collected feedback. After completing the tasks, participants filled out a System Usability Scale (SUS) survey and provided satisfaction ratings for each module.

The study goals align closely with user-centered design principles because they focus on understanding the needs and experiences of actual users. User-centered design emphasizes designing systems around users rather than expecting users to adapt to the system. Garofalo (2025) explains that usability testing allows designers to identify barriers, reduce user frustration, and improve overall effectiveness. By observing participants as they completed realistic library tasks, researchers were able to gather valuable information about areas where students struggled and areas where the onboarding document successfully supported users. The findings from the study can inform future design decisions. For example, sections with low task completion rates may require clearer instructions, updated screenshots, or additional explanations. Areas with high error rates may need better guidance and navigation support. Also, user feedback can help designers determine where terminology should be simplified and where additional training materials may be beneficial. Garofalo (2025) notes that organized reporting and action plans are essential because they help teams transform usability findings into meaningful design improvements.

Description of the Interface

The interface being evaluated was LibraryStart, an informational onboarding document developed to help new students learn how to use the university library's digital resources. The document contains written instructions, screenshots, and step-by-step guidance for completing common library tasks. Students reviewed the onboarding document before attempting to complete tasks using the actual library website and online systems. Several design elements influence the usability of the onboarding document. These include the clarity of instructions, the quality of screenshots, the consistency of terminology, and the overall organization of information. The document serves as a guide that helps users move between instructional content

and the live library portal. Therefore, the effectiveness of the document depends on how clearly it communicates information and how easily users can apply the instructions to the real system.

Gestalt principles also play a key role in how users interpret and understand information.

According to Garofalo (2025), users naturally group related elements together and rely on visual organizations to understand relationships between different pieces of information. Effective use of visual hierarchy, grouping, and proximity can make instructions easier to follow and reduce cognitive effort. Since LibraryStart relies heavily on screenshots and visual guidance, these principles directly affect usability.

Core User Workflows Tested

The study evaluated three primary user workflows:

The first workflow focused on searching for the library catalog. Participants were asked to locate a print book, find and open an e-book, and save an item to a reading list. These tasks measured how effectively students could search for and organize library resources. The second workflow involved accessing research databases. Participants navigated from the library homepage to a psychology database, located in a peer-reviewed article, and downloaded a full-text PDF. These tasks evaluated users' ability to perform academic research using the library's online resources.

The third workflow focused on interlibrary loan requests. Participants searched for an unavailable book, submitted an interlibrary loan request, and attempted to locate the status page for their request. These tasks measured how effectively users could obtain materials that were not available directly through the library's collection.

Interface Strengths and Weaknesses

One of the strongest areas identified in the study was the catalog search module. Task success rates ranged from 88% to 96%, and participants completed tasks quickly with very few errors. Satisfaction ratings were also high, with students reporting strong levels of clarity, helpfulness, and confidence. Participant comments further support these positive results. Several users stated that the screenshots matched what they saw on the screen, and that icons were easy to understand. One participant noted that the reading list icon resembled a bookmark, making its purpose immediately clear. Another participant was able to find the necessary filters quickly and complete tasks without difficulty. These strengths align with Nielsen's heuristic recognition rather than recall. Users were able to recognize important interface elements because the onboarding document provided detailed screenshots and visual cues. The heuristic evaluation also noted that Module 1 included annotated screenshots and a common mistakes section that helped prevent user errors. The research database workflow demonstrated moderate success but also revealed several weaknesses. Success rates decreased to between 79% and 87%, while completion times and error counts increased significantly. Many participants struggled to locate database links and identify peer-reviewed articles. Some users became confused because the database interfaces did not match the screenshots shown in the onboarding document. Others did not fully understand the meaning of "peer-reviewed" and had trouble using the available filters. These findings suggest problems with consistency and terminology. Nielsen's usability heuristics emphasize the importance of consistency and standards because users should not have to interpret different terms that refer to the same concept. The heuristic evaluation specifically noted inconsistencies in the use of terms such as "catalog," "search," "databases," and "research resources."

The weakest area of the onboarding document was the interlibrary loan workflow. Task success rates ranged from only 54% to 71%, making it the lowest-performing section of the study. Participants experienced the highest error rates and required far more clicks than expected to complete tasks successfully. Task 3.2, submitting an interlibrary loan request, had the lowest success rate and highest number of errors among all nine tasks. User observations revealed several significant issues. Participants struggled to locate the interlibrary loan option, became confused by separate login requirements, missed important form validation messages, and could not easily locate the status page for submitted requests. One participant commented that the instructions contained too much text but did not clearly explain the required actions. These weaknesses relate directly to Nielsen's heuristics of visibility of system status and error prevention. Users were not provided with sufficient feedback regarding their progress, and the onboarding document did not adequately prepare them for common issues such as authentication requirements. The heuristic evaluation also found that the ILL section lacked detailed screenshots, troubleshooting information, and examples showing what successful completion looked like. The implications of these weaknesses are important because students who encounter difficulties may become frustrated and avoid using valuable library services. Improving these workflows could increase efficiency, reduce errors, and improve overall user confidence.

Overall Effectiveness of the Workflows

Overall, the LibraryStart onboarding document demonstrated moderate effectiveness. The catalog search workflow performed very well and showed that students could successfully complete basic library tasks after reviewing the document. However, the research database and interlibrary loan workflows revealed several usability concerns that negatively impacted performance and user satisfaction.

The System Usability Scale score for the study was 64.2, which is below the industry average score of 68. This suggests that while participants found the system usable, there is significant room for improvement. Satisfaction ratings also declined from Module 1 to 3, indicating that users had increased difficulty as tasks became more complex. The results demonstrate the value of usability testing and structured reporting. Garofalo (2025) explains that usability reports help organizations prioritize issues and develop action plans that support future design improvements. Based on the findings, the library should focus on updating screenshots, simplifying terminology, improving navigation guidance, and providing additional support for interlibrary loan tasks.

Conclusion

The LibraryStart usability study provided valuable insights into how incoming students interact with the university library's online resources. The study identified impressive performance in catalog searching while revealing significant challenges in research database navigation and interlibrary loan requests. Evidence from task completion rates, user feedback, satisfaction ratings, and heuristic evaluations clearly shows where improvements are needed. Applying user-centered design principles, usability heuristics, and effective reporting practices, the library can make targeted improvements that enhance usability, increase user confidence, and create a better experience for future students.

No AI tools were used.

References

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